

Face Recognition with Logistic Regression and PCA

Identifying faces from the Olivetti dataset using scikit-learn and PyTorch logistic regression, with PCA-based dimensionality reduction.

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Overview

Face recognition is a classic multi-class classification problem that tests a model's ability to distinguish fine-grained visual patterns. This project applies Logistic Regression to the Olivetti faces dataset (400 grayscale images of 40 individuals, each represented as a 4096-dimensional vector) using two frameworks: scikit-learn and PyTorch. PCA is used to explore how dimensionality reduction affects recognition accuracy.

Goals

Build and evaluate logistic regression classifiers for face recognition, compare scikit-learn and PyTorch implementations, and analyse how architecture choices (hidden layers, optimizer, PCA components) affect accuracy on a held-out test set.

Objectives

- Load and prepare the Olivetti faces dataset with a 70/30 stratified train/test split.
- Train a scikit-learn Logistic Regression model with 10-fold cross-validation and evaluate on the test set.
- Implement an equivalent model in PyTorch as a single linear layer (4096 to 40), and compare performance to scikit-learn.
- Analyse accuracy vs the number of hidden layers and vs the choice of optimizer (SGD, Adam, RMSprop, Adagrad).
- Apply PCA and plot Logistic Regression accuracy as the number of principal components increases from 2 to 10.

Approach

Data is loaded via sklearn's `fetch_olivetti_faces` and split into training and test sets. The scikit-learn pipeline uses KFold cross-validation to estimate generalisation performance before final test evaluation. The PyTorch pipeline standardises features with `StandardScaler` (fit on the training set only), then trains models with varying architectures and optimizers. PCA experiments reduce the feature space before fitting a logistic regression model, measuring accuracy at each component count.

Outcomes

- Scikit-learn Logistic Regression achieved 10-fold CV accuracy of 0.939 and test accuracy of 0.975.
- PyTorch Logistic Regression achieved 10-fold CV accuracy of 0.932 and test accuracy of 0.958.
- Adding hidden layers (up to 3 layers) did not improve test accuracy; the plain linear model performed best on this small dataset.

- PCA accuracy on the test set increased from 0.15 with 2 components to 0.88 with 10 components, showing strong dependence on the number of retained dimensions.